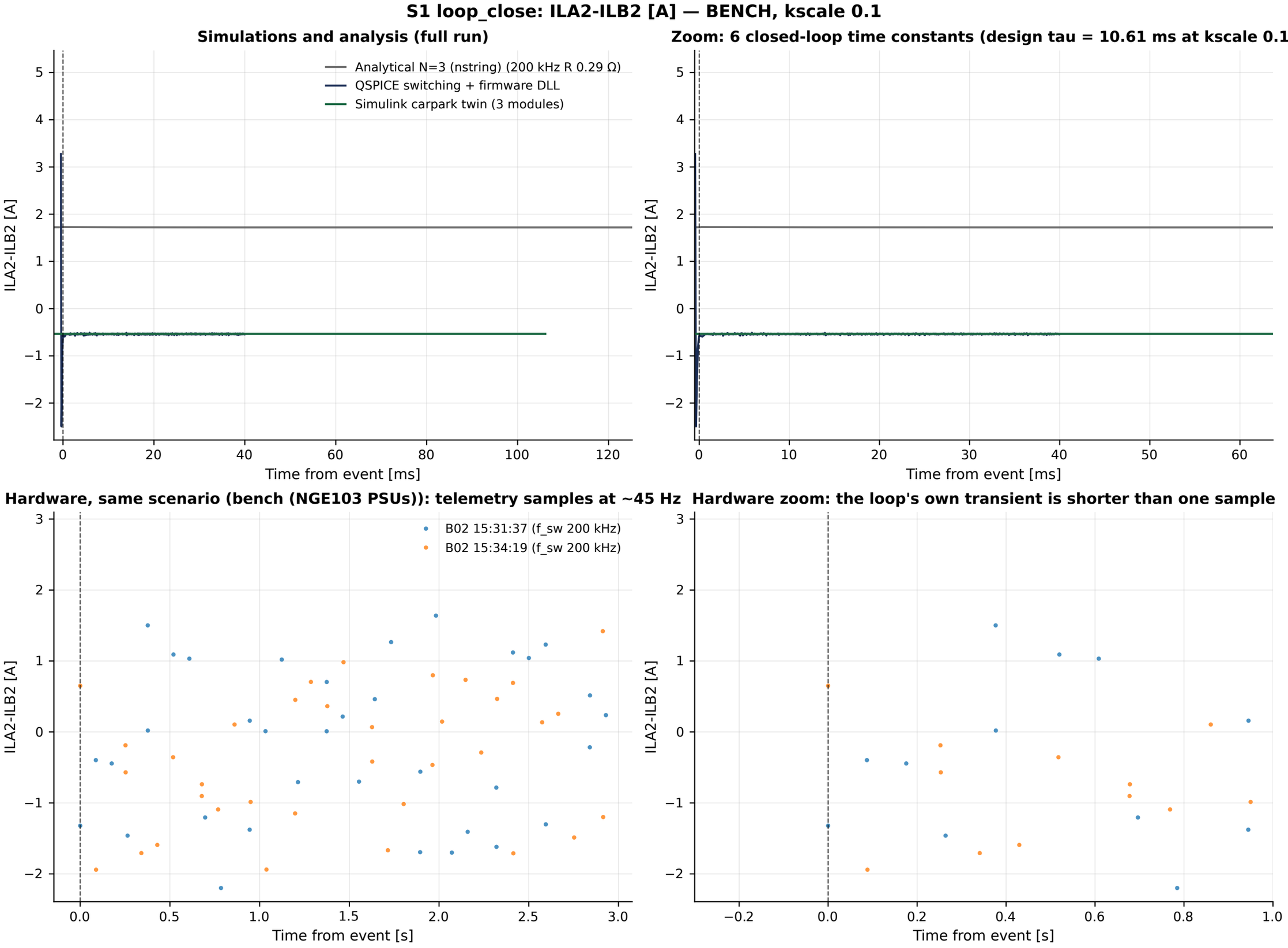
S1 loop_close: ILA2-ILB2 [A] — BENCH, kscale 0.1
Simulations and analysis (full run)
Analytical N=3 (nstring) (200 kHz R 0.29 Ω)
QSPICE switching + firmware DLL
Simulink carpark twin (3 modules)
ILA2-ILB2 [A]
Time from event [ms]
Zoom: 6 closed-loop time constants (design tau = 10.61 ms at kscale 0.1
ILA2-ILB2 [A]
Time from event [ms]
Hardware, same scenario (bench (NGE103 PSUs)): telemetry samples at ~45 Hz
B02 15:31:37 (f_sw 200 kHz)
B02 15:34:19 (f_sw 200 kHz)
ILA2-ILB2 [A]
Time from event [s]
Hardware zoom: the loop's own transient is shorter than one sample
ILA2-ILB2 [A]
Time from event [s]